Modeling of Thermal Systems

(热系统的建模)

Main Contents of Course (课程主要内容)

Mechanical (机械)

Electrical (电路)

Electro-Mechanical (机–电)

Fluid (流体)

Thermal (热系统)

Dynamic Modeling (动态建模)

Model structure (模型结构)

System Identification(系统辨识)

Parameter estimation (参数估计)

Control System Design(控制系统设计)

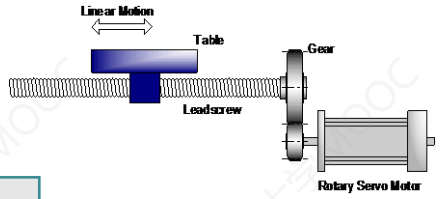
System Analysis (系统分析)

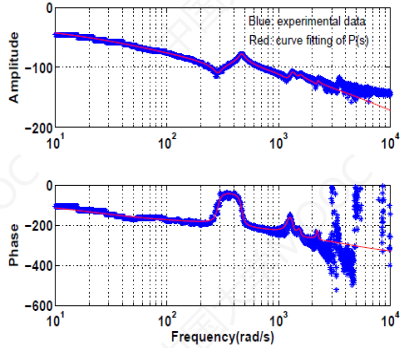
Root-locus design (根轨迹设计)

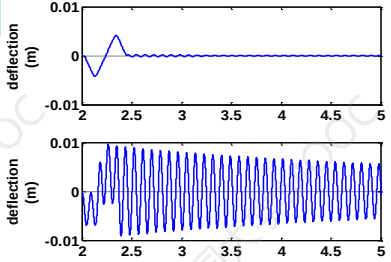
Pole placement design (极点配置)

Design limitations (性能局限)

Architectural issues (控制结构)







Variables

- ***q***: heat flow rate (热流量) **[J/sec = W]** (current (电流))
- ***T***: temperature (温度) **[^{o}K] or [^{o}C]** (voltage (电压))

Notes:

➢ Dynamics of a thermal system are usually described by partial differential equations since temperature in a body usually depends on spatial as well as temporal coordinates **(由于物体中温度通常依赖于空间和时间坐标，热系统的动力学通常由偏微分方程来描述)**.

➢ Nonlinearities are often essential in describing the heat transfer by radiation and convection **(经常需要用非线性表述辐射和对流热传递特性)**.

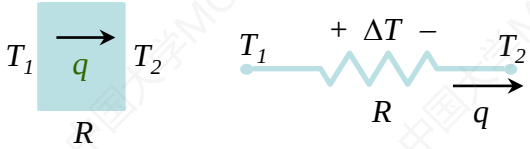
➢ Usually, finite element methods (FEM) are used to numerically solve nonlinear PDE problems since very few nonlinear PDEs have analytical (closed form) solutions **(由于偏微分方程极少能用解析解求解，通常用有限元方法进行非线性偏微分问题的数值求解)**.

➢ Lumped model approximations of thermal systems are used to obtain linear ODEs that are capable of describing the dynamic response of thermal systems to a good first approximation **(用热系统的集中模型近似获得线性常微分方程以描述热系统动态响应的一阶近似量)**.

Basic Modeling Elements

- **Thermal Resistance (热阻)**

  Describes the heat transfer process through an element with the characteristic that the heat flow rate across the element is proportional to the temperature difference across the element, i.e **(反映通过元件的热传递过程，表示经过元件的热流量对两端温度差的比值).**

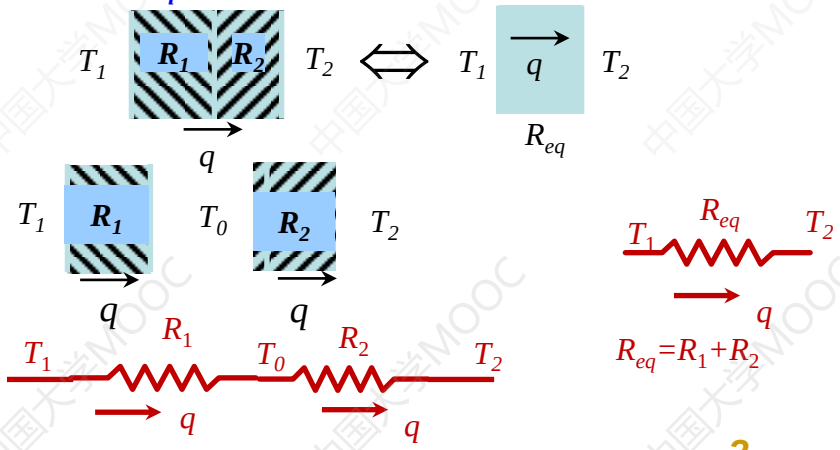


$$\Delta T = T_{12} = T_1 - T_2 = R \cdot q$$

or

$$q = \frac{1}{R}(T_1 - T_2) = \frac{1}{R}\Delta T$$

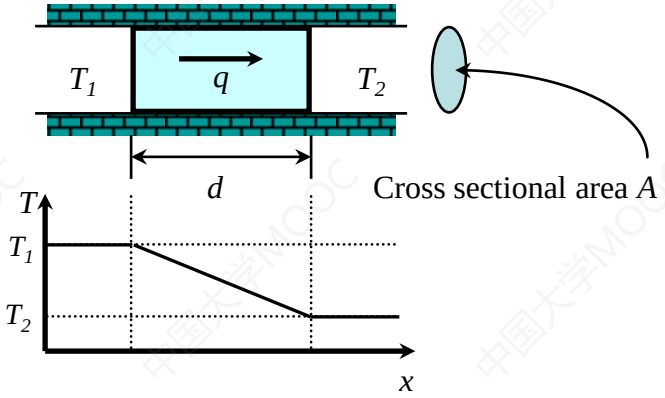
Ex: Two bodies at temperatures T_1 and T_2 are separated by two elements with different thermal resistance R_1 and R_2. Heat flows through the two elements at a rate of q. Find the equivalent thermal resistance R_{eq} and solve for the interface temperature between the two elements **(两温度为T_1和T_2的物体由两个热阻为R_1和R_2的元件隔离，流过两元件的热流量为q,找出等效热阻R_{eq}和求解两元件的界面温度).**



Three Types of Heat Transfer

- **Conduction (热传导)**

  Heat transfer through solid or continuous media via random molecular motion (diffusion) **(在固体或连续介质上通过随机分子运动进行的热传递).**



$$q = \frac{\alpha A}{d}(T_1 - T_2) = \frac{\alpha A}{d} T_{12} \quad \Rightarrow \quad R = \frac{d}{\alpha A}$$

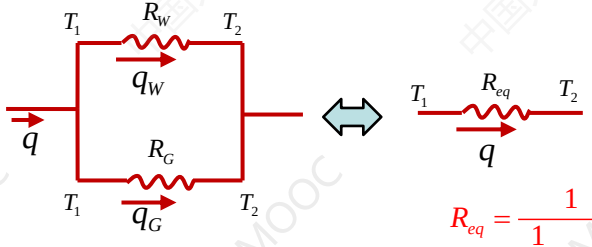
$$\Rightarrow T_{12} = \underbrace{\frac{d}{\alpha A}}_{R} q$$

– a : thermal **conductivity (导热系数)** [W/(m.°K)]

Ex: Calculate the equivalent thermal resistance of a wall with a **window (计算带有窗户的墙的等效热阻).**

	Wall	Window
Area	A_W	A_G
Thickness	d_W	d_G
a	a_W	a_G

$$R_W = \frac{d_w}{\alpha_w A_w} \qquad R_G = \frac{d_G}{\alpha_G A_G}$$

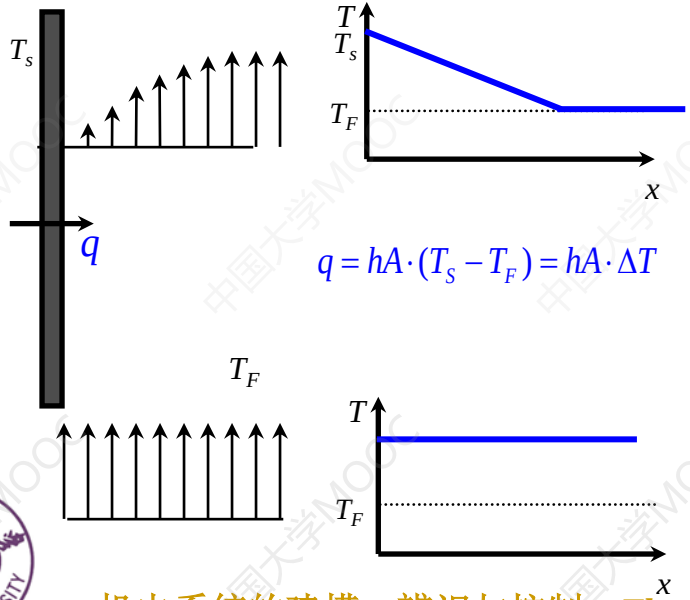


$$R_{eq} = \frac{1}{\frac{1}{R_w} + \frac{1}{R_G}}$$

Three Types of Heat Transfer

- **Convection** (对流)

  Heat transfer between the interface of a solid material and a fluid material via bulk motion of the fluid (在固体介质和流体介质上通过流体弹性运动进行的热传递).



$$q = hA \cdot (T_S - T_F) = hA \cdot \Delta T$$

$$\Rightarrow R = \frac{1}{hA}$$

– A : surface area (表面积) [m^2]

– h : convective heat transfer coefficient (对流传热系数) [W/m^2-^{o}K], which depends on surface geometry, fluid flow rate, temperature, flow direction, ... (依赖于表面几何形状，流量，温度，流动方向等)

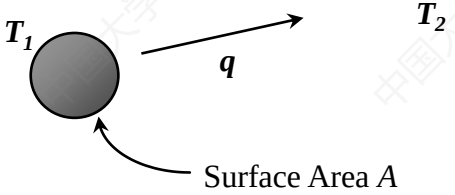
– T_S :surface temperature (表面温度) [^{o}K]

– T_F :fluid temperature (流体温度) [^{o}K]

Three Types of Heat Transfer

- **Radiation (热辐射)**

  Heat transfer via electromagnetic waves (通过电磁波进行热传递)



$$q = \sigma F_E F_V A \cdot (T_1^4 - T_2^4)$$

– A : surface area (表面积) [m^2]

– σ : Stefan-Boltzmann constant (史蒂芬-波兹曼常数) [W/(m^2.^{o}K^4)]

– F_E : effective emissivity (有效辐射系数)

– F_V : view factor (视系数)

Nonlinear! (非线性)

***Note*:(注意点)**

➢ Except for radiation, both conductive and convective heat transfer processes can be modeled as thermal resistances (除了热辐射，热传导和热对流的传热过程都可用热阻来建模).

➢ In the previous discussions, the assumption is that the materials do not store thermal energy. In reality, materials do store a certain amount of thermal energy (以上讨论中，假设介质没有储存热量。实际上，介质可以储存一定的热量).

Q: How would we model the process of storing thermal energy **? (问：如果介质储存热能量的过程是怎么建模呢？)**

Basic Modeling Elements

- **Thermal Capacitance (热容)**

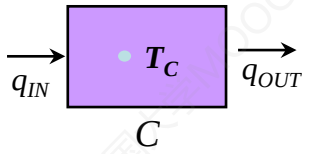
The ability of a substance to hold or store heat is the heat capacity of the material and it behaves like a thermal capacitance. Since the specific heat c_P can be interpreted as the heat storage capacity of the material per unit mass, the total heat storage capacity of a material is (物体维持或存贮热量的能力是介质的热能力，可看做热容。由于比热可解释成单位质量介质的热存储能力，一个介质的全部热存贮能力表示为)：

$$c_P \cdot M$$

If there is net heat flow into the material, the temperature of the material will change and the rate of temperature change is proportional to the net heat flow rate q_{store}: (如果有净热流量流入介质，介质的温度将会改变，并且温度变化率正比于净热流量q_{store})

$$c_P M \frac{d}{dt} T_C = q_{store} = q_{IN} - q_{OUT}$$

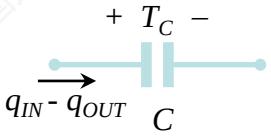
Thermal Capacitance (热容)



Mass, M
Volume, V
Density, ρ

$$C = c_P M = c_P \rho V$$

Equivalent Electric Element
(等效电路元件)



Note: The above relationship holds only if we assume that the temperature is uniform across the entire material. (注意：上面的关系仅在我们假设整个介质上的温度是均匀的)

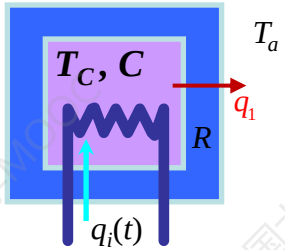
Interconnection Laws

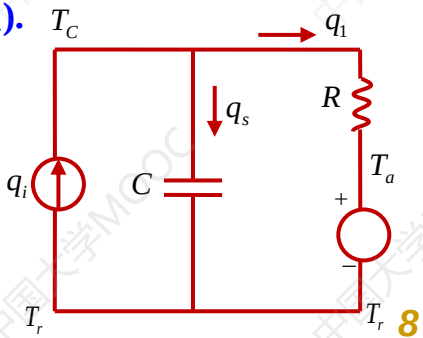
- **Energy Balance - 1st Law of Thermodynamics (能量平衡-热力学第一定律)**
  - Energy stored in the system is the sum of the net energy inflow, the energy generated within the system and the work done on the system **(系统中存贮的能量是净输入流量、系统内产生的能量和对系统做的功的总和):**

$$q_{store} = \sum q_{IN} - \sum q_{OUT} + q_{\substack{GENERATED' \\ WITHIN}} + \frac{d}{dt} W_{\substack{WORK \\ DONE}}$$

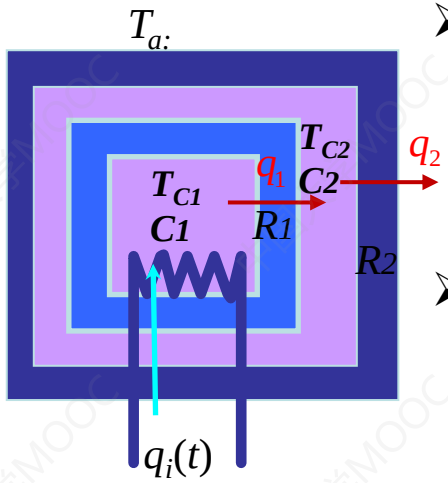
  - Note that there is ***NO inertia*** in macroscopic thermal system **(注意宏观热系统没有惯性)**

Ex: A material with a thermal capacitance C is surrounded by an insulation material with thermal resistance R. Heat is added to the inner material at a rate of $q_i(t)$. Find the system model, if the inner material temperature T_C is to be the output **(热容是C的介质被热阻是R的绝缘材料包围。有一热流量$q_i(t)$被加入到内部介质中。找出以内部介质温度T_C为输出量的系统模型).**





Example (a refined model)

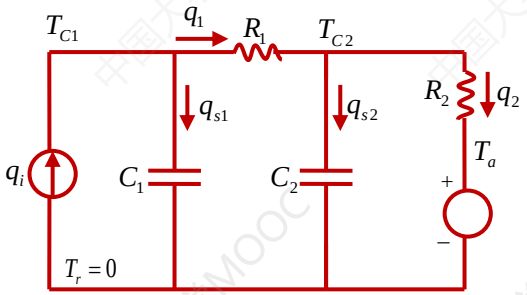


➢ ***Element Eqs* (元件方程):**

$$q_{s1} = C_1\dot{T}_{c1}$$
$$T_{c1} - T_{c2} = R_1 q_1$$
$$q_{s2} = C_2\dot{T}_{c2}$$
$$T_{c2} - T_a = q_2 R_2$$

➢ ***Interconnection* (互联关系):**

$$q_i = q_{s1} + q_1$$
$$q_1 = q_{s2} + q_2$$



➢ ***Uvs* (未知变量):**

$q_{s1}, T_{c1}, T_{c2}, q_1, q_{s2}, q_2$ 6 UVs=6 eqs

➢ ***Inputs* (输入量):**

q_i, T_a

➢ ***EOMs* (运动方程):**

$$\begin{cases} C_1\dot{T}_{c1} + \dfrac{1}{R_1}T_{c1} - \dfrac{1}{R_1}T_{c2} = q_i \\ C_2\dot{T}_{c2} + (\dfrac{1}{R_1} + \dfrac{1}{R_2})T_{c2} - \dfrac{1}{R_1}T_{c1} = \dfrac{1}{R_2}T_a \end{cases}$$

➢ ***If T_{c1} is the output, the I/O model is* (T_{c1} 为输出量的输入/输出模型):**

$$\left\{C_1C_2s^2 + (\frac{C_1}{R_{eq}} + \frac{C_2}{R_1})s + \frac{1}{R_1R_2}\right\}T_{c1}(s) = (C_2s + \frac{1}{R_{eq}})Q_i(s) + \frac{1}{R_1R_2}T_a(s)$$

$$\Rightarrow C_1C_2\ddot{T}_{c1} + (\frac{C_1}{R_{eq}} + \frac{C_2}{R_1})\dot{T}_{c1} + \frac{1}{R_1R_2}T_{c1} = C_2\dot{q}_i + \frac{1}{R_{eq}}q_i + \frac{1}{R_1R_2}T_a$$

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2}$$

In Class Example

The Pentium processor under normal operation will generate heat at a rate of $q_i(t)$. **(正常工作的奔腾处理器会产生热流量)**

The processor itself has a specific heat of c_P.

The cross sectional area of the chip is A_P with a thickness of d_P.

The average density of the processor is ρ_P .

To help dissipate the heat and reduce the processor temperature T_P, a heat sink with the same cross sectional area and an average thickness of d_S is added on top of the processor. **(为帮助散热，降低处理器温度T_P,在处理器上增加了1个散热片)**

The heat sink has a thermal conductivity of α_S .

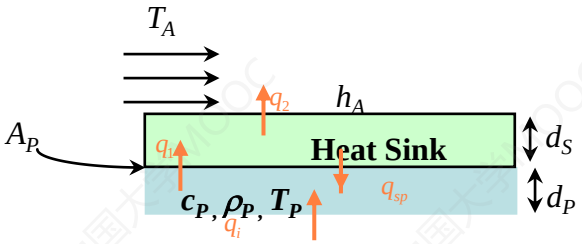
The effective conduction area between the heat sink and processor is approximated by A_P .

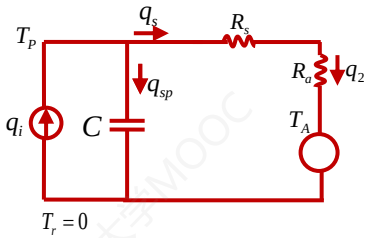
To further improve heat dissipation, a fan is used to generate air flow on top of the heat sink **(为进一步提高散热能力，在散热片上面用一个风扇产生空气流动)**,

The effective convection coefficient is h_A and the effective contact area between the heat sink and the air flow is A_S .

The temperature of the fan air is maintained at T_A.

Find the relationship between $q_i(t)$ and the temperature of the processor T_P.**(请找出q_i和处理器温度T_P之间的关系)**





$$R_s = \frac{d_s}{\alpha_s A_p}$$

$$R_a = \frac{1}{h_A A_s}$$

$$C = c_p \rho_p A_p d_p$$

$$R_{eq} = R_s + R_a$$

$$q_{sp} = C\dot{T}_p$$

$$T_p - T_A = R_{eq} q_s$$

$$q_i = q_{sp} + q_s = q_{sp} + q_2$$

$A_p \uparrow, A_s \uparrow, d_s \downarrow$

$\Rightarrow R_{eq} \downarrow, T_p(0) \downarrow$ with same q_i

$\Rightarrow$

$$R_{eq} C\dot{T}_p + T_p = R_{eq} q_i + T_A$$

$$T_p(s) = \frac{R_{eq} Q_i(s)}{R_{eq} Cs + 1} + \frac{T_A(s)}{R_{eq} Cs + 1},$$

Summary

- **Basic Modeling Elements (基本建模元素)**
  - Resistance (热阻)
    - Conduction (热传导)
    - Convection (热对流)
    - Radiation (热辐射)
  - Capacitance (热容)
- **Interconnection Relationships (互联关系)**
  - Energy Balance - 1st Law of Thermodynamics (能量平衡-热力学第一定律)
- **Derive Input/Output Models (推导输入/输出模型)**